



Description
No Data

Simulation of auger and shaft

Date: Tuesday, January 6, 2026
Designer: Solidworks
Study name: auger and shaft analysis
Analysis type: Static
ENG name:Hassan Moqbel Morshed

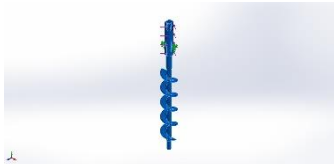
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Assumptions

Model Information

 Model name: auger and shaft Current Configuration: Default			
Solid Bodies			
Document Name and Reference	Treated As	Volumetric Properties	Document Path/Date Modified
Combine1 	Solid Body	Mass:1.66362 kg Volume:0.000207953 m^3 Density:8,000.01 kg/m^3 Weight:16.3035 N	C:\Users\user1\Desktop\VFFS\auger and shaft analysis\auger and shaft.SLDPRT Jan 6 02:47:55 2026

Study Properties

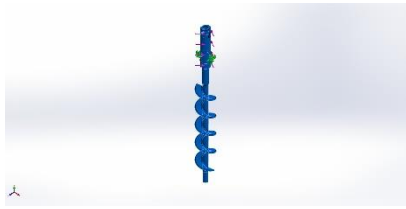
Study name	auger and shaft analysis
Analysis type	Static
Mesh type	Solid Mesh
Thermal Effect:	On
Thermal option	Include temperature loads
Zero strain temperature	298 Kelvin
Include fluid pressure effects from SOLIDWORKS Flow Simulation	Off
Solver type	Automatic
Inplane Effect:	Off
Soft Spring:	Off
Inertial Relief:	Off
Incompatible bonding options	Automatic
Large displacement	Off
Compute free body forces	On
Friction	Off
Use Adaptive Method:	Off
Result folder	SOLIDWORKS document (C:\Users\user1\Desktop\VFFS\auger and shaft analysis)

Units

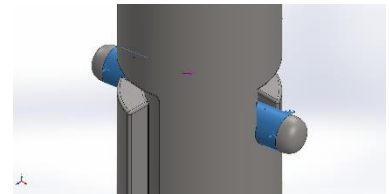
Unit system:	SI (MKS)
Length/Displacement	mm
Temperature	Kelvin
Angular velocity	Rad/sec
Pressure/Stress	N/m ²

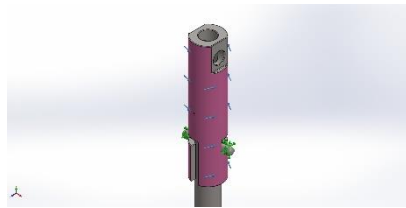


Material Properties

Model Reference	Properties	Components
	Name: AISI 304 Model type: Linear Elastic Isotropic Default failure criterion: Max von Mises Stress Yield strength: 2.06807e+08 N/m ² Tensile strength: 5.17017e+08 N/m ² Elastic modulus: 1.9e+11 N/m ² Poisson's ratio: 0.29 Mass density: 8,000 kg/m ³ Shear modulus: 7.5e+10 N/m ² Thermal expansion coefficient: 1.8e-05 /Kelvin	SolidBody 1(Combine1)(auger and shaft)
Curve Data:N/A		

Loads and Fixtures

Fixture name	Fixture Image	Fixture Details		
Fixed-1		Entities: 2 face(s) Type: Fixed Geometry		
Resultant Forces				
Components	X	Y	Z	Resultant
Reaction force(N)	-0.976147	-0.0241341	-0.00671291	0.976468
Reaction Moment(N.m)	0	0	0	0

Load name	Load Image	Load Details
Torque-1		Reference: Face< 1 > Type: Apply torque Value: -20 N.m



Connector Definitions

No Data

Interaction Information

No Data

Mesh information

Mesh type	Solid Mesh
Mesher Used:	Blended curvature-based mesh
Jacobian points for High quality mesh	16 Points
Maximum element size	6.22279 mm
Minimum element size	0.311139 mm
Mesh Quality	High

Mesh information - Details

Total Nodes	535005
Total Elements	351350
Maximum Aspect Ratio	64.285
% of elements with Aspect Ratio < 3	99.3
Percentage of elements with Aspect Ratio > 10	0.0111
Percentage of distorted elements	0
Time to complete mesh(hh:mm:ss):	00:00:39
Computer name:	

Sensor Details

No Data



Resultant Forces

Reaction forces

Selection set	Units	Sum X	Sum Y	Sum Z	Resultant
Entire Model	N	-0.976147	-0.0241341	-0.00671291	0.976468

Reaction Moments

Selection set	Units	Sum X	Sum Y	Sum Z	Resultant
Entire Model	N.m	0	0	0	0

Free body forces

Selection set	Units	Sum X	Sum Y	Sum Z	Resultant
Entire Model	N	-0.0741707	-0.142409	-0.307086	0.346531

Free body moments

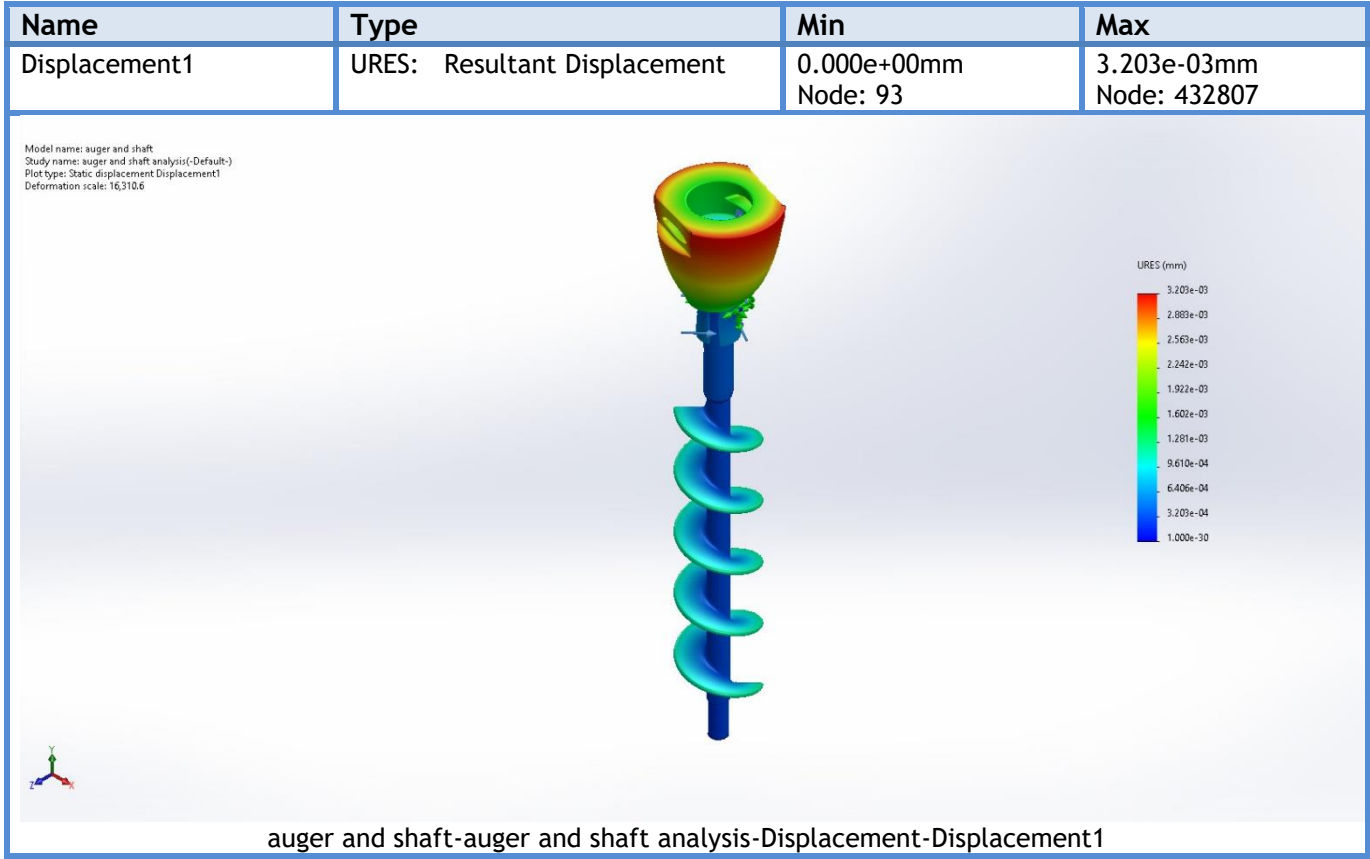
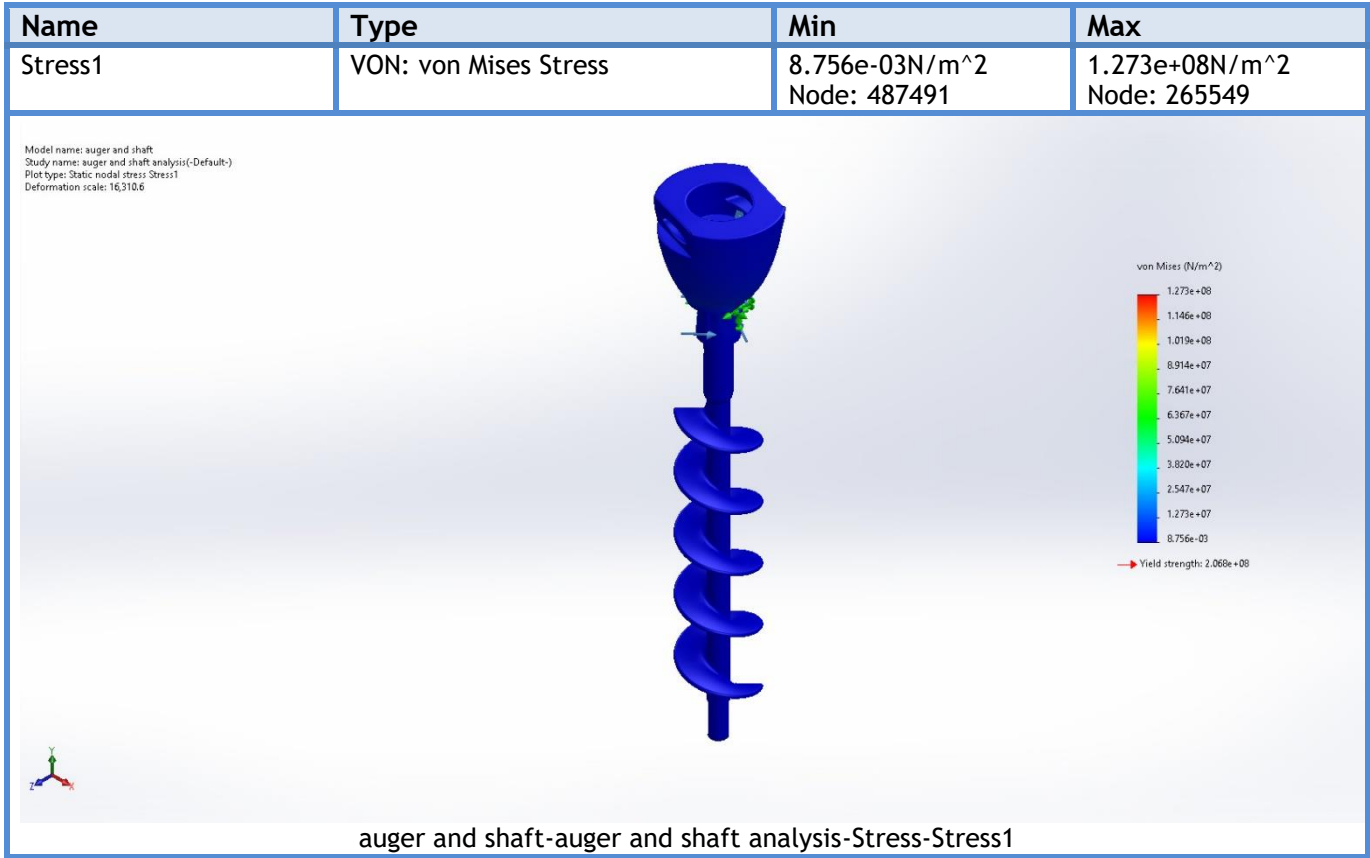
Selection set	Units	Sum X	Sum Y	Sum Z	Resultant
Entire Model	N.m	0	0	0	1e-33

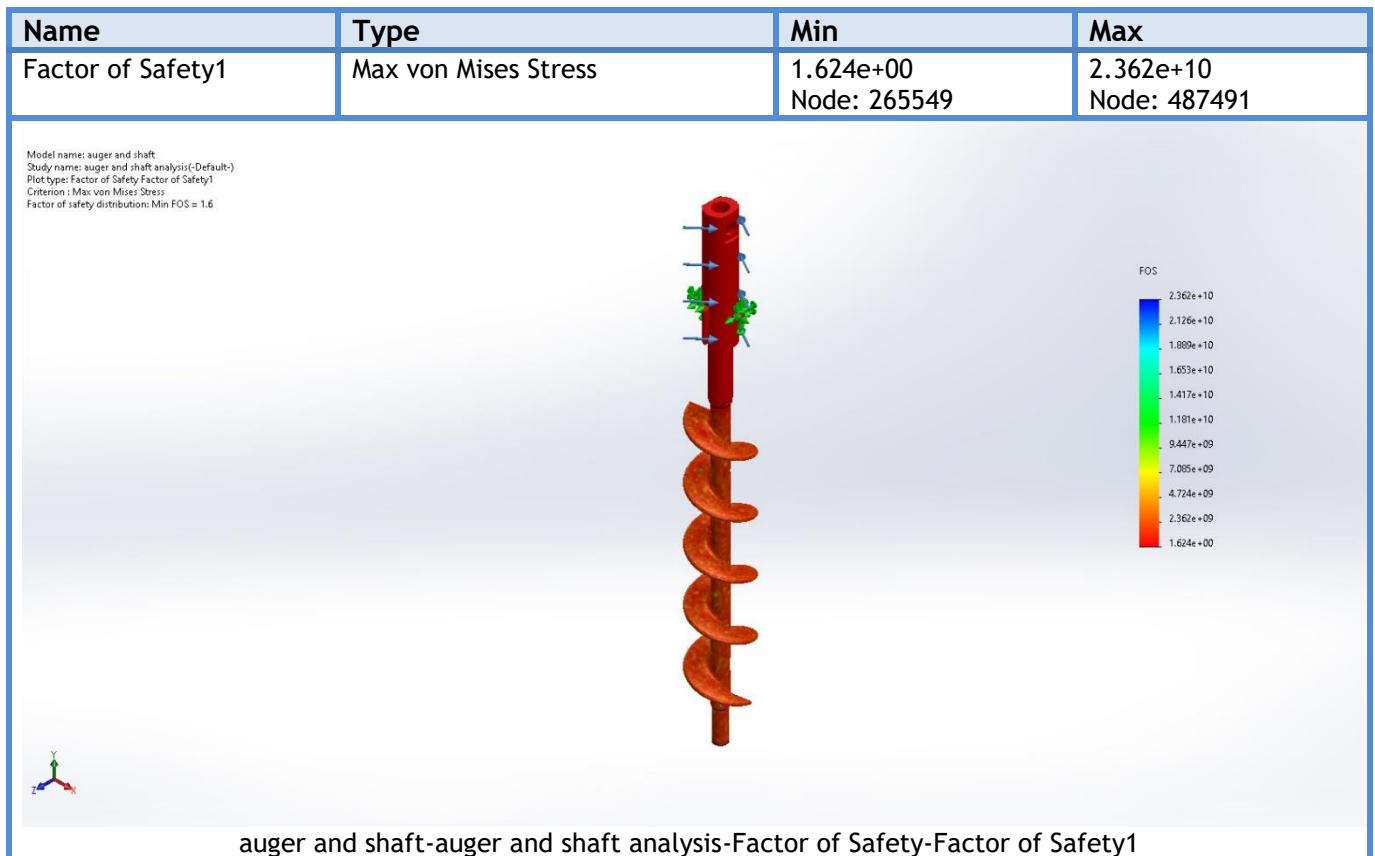
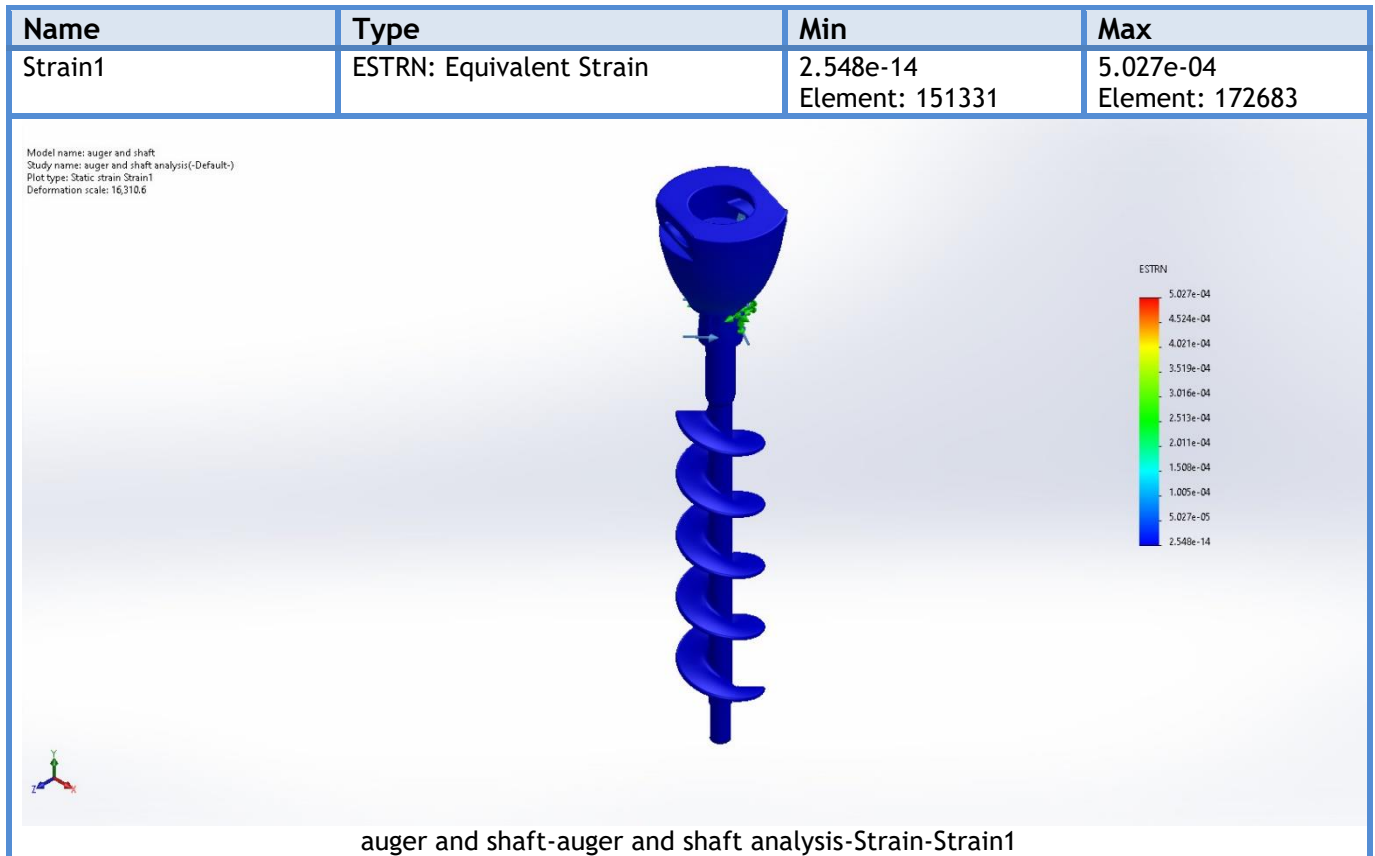
Beams

No Data



Study Results





Conclusion

